

CHAPTER 4 (PART I): INTELLECTUAL PROPERTY

**TS. DR. ROZI NOR HAIZAN
BINTI NOR**

OBJECTIVES

- What is IP and how new technologies affect IP
- How should IP be protected
- Copyright law
- Free software
- Issues for software developers



INTRODUCTION

- Digital music storage + Internet --> crisis
- Value of intellectual properties much greater than value of media
 - Creating first copy is costly
 - Duplicates cost almost nothing
- Illegal copying pervasive
- In light of information technologies, how should we treat IP?

DEFINITION - IP

- Any unique product of the human intellect that has commercial value
- Intangible creative work – not necessarily the physical form on which it is stored or delivered
 - Books, songs, movies
 - Paintings, drawings
 - Inventions, chemical formulas, computer programs
- IP ≠ physical manifestation

INTELLECTUAL PROPERTY

- Intellectual property includes works of the mind such as art, books, films, formulae, inventions, music, and processes that are unique and that are “owned” or created by a single entity.



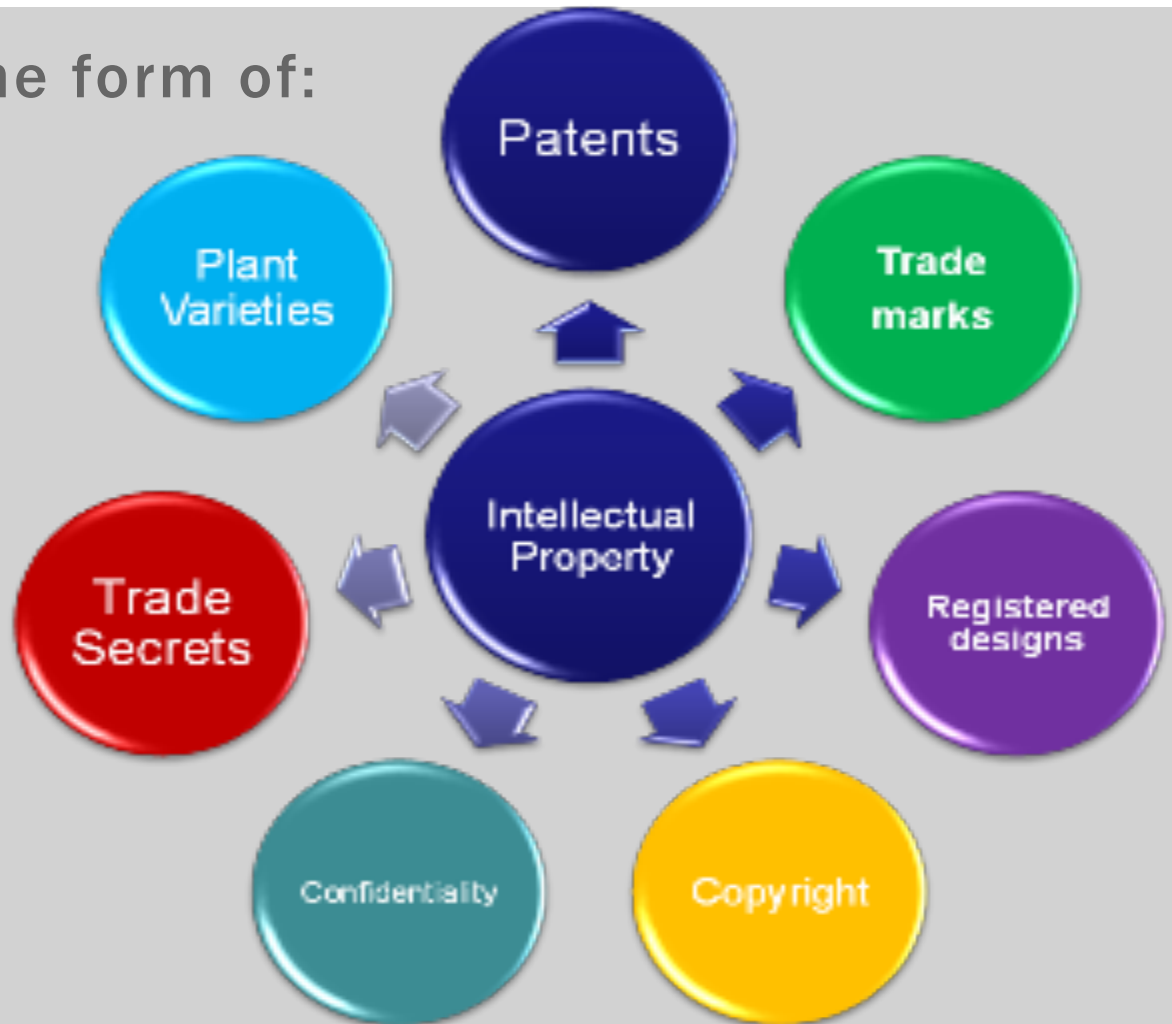
ANALOGY IS IMPERFECT

- If two people write same play, both cannot own it
 - every IP is one-of-a-kind
- If one person “takes” another's play, both have it
 - copying an IP is different from stealing a physical object

LEGAL PROTECTION

■ IPs are given in the form of:

- Trade secret
- Patents
- Trademark
- copyright



WHY LEGAL PROTECTION?

- The value of IP comes from the creativity, ideas, research, skills, labour, and other nonmaterial effort and attributes provided by the creators
- A property right is a type of 'natural right' that should be granted
- Property rights are social contracts designed to encourage creators and inventors to better serve society

BENEFITS OF IP PROTECTION

- New ideas in the form of inventions and artistic works can improve the quality of life
- Some people are altruistic and glad to share with others; some are not
- attraction of wealth can be the incentive to produce something useful
- If a person has a right to control the distribution and use of a piece of IP, there are opportunities to make money
- Government recognized benefits to limited intellectual property protection

LIMITS TO IP PROTECTION

- Giving inventors rights to their inventions stimulates creativity
- Society benefits most when inventions in public domain
- Inventors or authors are given rights for a limited time

PROTECTING IP – TRADE SECRETS

- The Uniform Trade Secrets Act (UTSA) defines a trade secret as “information, including a formula, pattern, compilation, program, device, method, technique, or process that:
 1. Derives independent economic value, actual or potential, from not being generally known to, and not being readily ascertainable by persons who can obtain economic value from its disclosure or use
 2. Is the subject of efforts that are reasonable under the circumstances to maintain its secrecy

PROTECTING IP – TRADE SECRETS

- Confidential piece of IP that gives company a competitive advantage
- Never expires
- Not appropriate for all IP
- May be compromised when employees leave firm

TRADE SECRETS

- Reverse Engineering Allowed
- Reverse engineering is the process of breaking something down in order to understand it, build a copy of it, or improve it.

TRADE SECRET

- To qualify as a trade secret, the secret must have
 1. Economic value
 2. Must not be readily ascertainable
 3. The trade secret owner must take steps to maintain its secrecy

PROTECTING IP – TRADEMARK, SERVICE MARK

- Trademark : identifies goods
- Service mark : identifies services
- Company can establish a “brand name”
- Does not expire
- If brand name becomes common noun, trademark may be lost



TM ®
© SM

- A trademark is anything that enables a consumer to differentiate one company's products from another
- TRADEMARK is something you can do to brands. If you trademark a brand, then you own the "intellectual property" of that brand and you are the only person allowed to use that Brand name, slogan etc.
If others want to use that brand, they must ask your permission or pay some money.

PROTECTING IP - PATENT

- A public document that provides detailed description of invention
- Provides owner with exclusive right to the invention
- Owner can prevent others from making, using, or selling invention for certain amount of years



PATENTS

- A patent enables the inventor to take legal action against those who, without the inventor's permission, manufacture, use or sell the invention during the period of time the patent is in force.
- The United States Patent and Trademark Office reviews patent applications and issues patents to approved inventions



COPYRIGHT

- Provides owner of an original work five rights
 - Make copies of the work
 - Distribute copies
 - Produce derivative works
 - Perform the work in the public
 - Display the work in the public



PROBLEMS FROM NEW TECHNOLOGIES

- High-quality copying
- High-quality distribution
- Easier to use
- Less expensive

PROBLEMS FROM NEW TECHNOLOGIES – TECHNOLOGICAL FACTORS

- Storage of all sorts of information in standard digitized formats
- High-volume, relatively inexpensive digital storage media (hard disks, CDRoms, DVD, thumb drive)
- Compression formats (MP3) make music and movie files small enough to download, copy, and store
- Character scanners and image scanners convert printed text, photos, and artwork to digitized forms

PROBLEMS FROM NEW TECHNOLOGIES

– TECHNOLOGICAL FACTORS

- The ease of “perfect” copying digitized material
- The ease of distributing digitized material over computer networks
- WWW makes it easy to find and download material
- Peer-to-peer technology permits easy transfer of files by large numbers of strangers over the internet without any centralized system or service

COPYRIGHT ACT 1987

- Protects copyright in Malaysia
- Provides characteristics of copyright material, scope of protection, and ways this protection will be given
- Provides clauses on enforcement

HISTORY OF COPYRIGHT IN MALAYSIA

- 1911 : enforcement of Copyright Act (UK) for Penang and Melaka
- 1930s : Copyright Enactment was used in Negeri Melayu Bersekutu.
- 1957 : Copyright Act 1911 and Copyright Enactment
- 1963 : Copyright Act 1956(UK) was a part of copyright laws in malaysia
- 1969 : Copyright Act 1969 was used replacing the earlier act
- 1987 : review of Copyright Act (1969) and used as Copyright Act (1987)
- 1990 : amendment to Copyright Act (1987)
- Until now there are amendments to Copyright Act (1987)

CONDITIONS FOR COPYRIGHT PROTECTION

- The work must be in one of the types mentioned in Copyright Act (1987)
- The work must be original
- The work should be fixed in material form

CONDITIONS FOR COPYRIGHT PROTECTION

- The inventor of the work should be qualified:-
 - citizen or permanent resident
 - firstly produced in Malaysia
 - made in Malaysia
 - made by or under the control of Malaysian government
 - citizen or permanent resident of a country in Treaty or Convention

COPYRIGHT INFRINGEMENT

- Direct Infringement
 - unauthorized person performs the exclusive rights of the owner
- Indirect Infringement
 - importing an infringed article from oversea for commercial without an agreement from the owner of the original copy, for example, if your brother has made infringing copies of a CD, and you know these copies are infringing and help create an e-bay listing for your brother to sell the discs, you have committed indirect infringement.
- Infringement through overriding of effective technology step
 - anyone does or makes others to do overriding to whatever technological steps used by the owner

COPYRIGHT INFRINGEMENT

- Infringement through alteration or deletion of electronic right management
 - the act of altering or deleting information on electronic right management without agreement from the owner

Electronic Right Management (ERM) technologies attempt to control what you can and can't do with the media and hardware you've purchased.

FAIR USE CONCEPT

- Sometimes legal to reproduce a copyrighted work without permission
 - allows uses of copyrighted material that contribute to the creation of new work and do not significantly affect sales of the material
 - allows some research and educational uses as well as news reporting and critiquing
 - however for public use, there should be an acknowledgement to the owner

FAIR USE CONCEPT

- Courts consider four factors
- Purpose and character of use
 - an educational use is more likely to be permissible than a commercial use
- Nature of work
 - use of non-fiction is more likely to be permissible than use of fiction. Published works are preferred over unpublished works

FAIR USE CONCEPT – CONT..

- Courts consider four factors
- Amount of work being copied
 - brief excerpts are more likely to be permissible than entire chapters
- Affect on market for work
 - use of out-of-print material is more likely to be permissible than use of a readily available work.

FAIR USE : EXAMPLE 1

A professor puts a few journal articles on reserve in the library and makes them assigned reading for the class. Some students in the class complain that cannot get access to the articles because other students always seem to have them checked out. The professor scans them and posts them on his web site. The professor gives the students in the class the password they need to access the articles.

FAIR USE : EXAMPLE 1

- First factor : the purpose of the use
 - The purpose is strictly educational; hence this factor weighs in favor of fair use
- Second factor : the nature of work
 - the journal articles are nonfiction. This weighs in favor of fair use
- Third factor : the amount of material
 - The fact that the professor is copying entire articles rather than brief excerpts weighs against a ruling of fair use

FAIR USE : EXAMPLE 1

o fourth factor : effect on the market for journal sales

➤ If the journal issues containing these articles are no longer for sale, then the professor's action cannot affect the market.

➤ The professor took care to prevent people outside the class from accessing the articles.

➤ Overall, this factor appears to weigh in favor of fair use

✓ Three out of four factors weigh in favor of fair use

FAIR-USE CASES

- Sony vs Universal City Studios
- Sony introduces Betamax VCR (1975)
- People start time shifting TV shows
- Movie studios sue Sony for copyright infringements
- US supreme court evaluated the case in light of the four fair use factors

FAIR-USE CASES

- First factor: intended purpose of the copying
 - Purpose is private, not commercial, time shifting should be seen as fair use
- Second factor : nature of the copied work
 - Consumers who are time shifting are copying creative work. this would tend to weigh against a ruling of fair use
- Third factor : amount of material copied
 - Since a consumer copies the entire work, this weighs against a ruling of fair use

FAIR-USE CASES

- Fourth factor : the affect time shifting will have on the market
 - The studios were unable to demonstrate that time shifting had eroded the commercial value
 - The movie studios receive large fees from television stations in return for allowing their movies to be broadcast
 - Television stations can pay these large fees because they receive income from advertisers
 - Advertising rates depend upon the size of the audience

FAIR-USE CASES

- Fourth factor : the affect time shifting will have on the market
 - The larger the audience, the more a television station can charge an advertiser
 - Time shifting allows people to watch a show later
 - Hence VCR can actually increase the size of audience, since the audience size determines the fees studios receive to have their movies broadcast, it is not at all clear whether copying harms the studios
- ✓ Supreme court ruled (5-4) that time shifting is a fair use

FAIR-USE CASES

- Diamond introduces Rio (1998)
- People start shifting their music
- RIAA starts legal action against Diamond for violation of the Audio Home Recording Act
- US court of appeal, 9th circuit, affirms that space shifting is consistent with copyright law

FAIR-USE CASES

- Sega Enterprises Ltd vs Accolade Inc
- 1992: reverse engineering a complete program in order to produce new, creative work was ruled fair use

COPYING MUSIC

- Improved technology allows for easy, fast, cheap, and ubiquitous copying on the web
- Entrepreneurs create businesses to facilitate storing and sharing of music files.
- Many individuals set up free sites for music sharing
- Recording industry continues to fight unauthorized copying of music

COPYING MOVIES AND TV PROGRAMS

- Improved digital technologies and greater bandwidth on the net also allows for copying and transferring of movies and tv programs
- Business such as RecocdTV.com and Scour provided free services to facilitate copying of broadcast intellectual material
- Entertainment companies continue to fight unauthorized copying of their IP

COPYING BOOKS

- Improved technology allows for simple, quick, and cheap copying of books
- Counterfeiters of textbooks, novel, and other printed matter, profit by not paying publishers and/or authors for their IP
- Electronic books use encryption to reduce copying, but some e-book protection schemes have been cracked

COPYING SOFTWARE

- Improved digital technologies contribute to unauthorized (commercial and non-commercial) copying of software
- Individuals and whole businesses, continue to produce, transport, and sell (or give away) copies of software, manuals and supporting material
- Software industry organizations and companies battle software piracy

THE NAPSTER CASE

- Benefits of Napster (aside from being free):
 - Share music with other users; obtain individual songs from a CD; sample songs on a CD; access more songs; access commercially unavailable songs;
- Legal issues:
 - Was copying and distributing music through Napster within fair-use guidelines? If not, was Napster responsible for their actions?
- Court decision: Napster was guilty of encouraging and assisting copyright infringement

PROTECTION FOR SOFTWARE

- What is exactly copyrighted software?
- Copyright protects the expression of idea. Eg, implementation of a relational DBMS; not the concept of using relational database to store information
- Usually protect the object (executable) program, not the source program
- Typically the source code to a program is confidential which is a trade secret
- Protects the screen displays produced by the program as it executes

SOFTWARE PIRACY – VIOLATION OF SOFTWARE COPYRIGHT

- The act of theft through copying software, use or distribute software product without authorization
- Copying of software (and documentation) to give or sell
- Illegal copying business or organization for their own use
- Distribution of software over the internet
- Preloading software onto the hard disk of a computer being sold
- Intentionally put copyrighted software on web sites for others to download

SOFTWARE PIRACY – VIOLATION OF SOFTWARE COPYRIGHT

- Counterfeiting – to produce, transport, and sell copies of the disks, documentation, and sometimes identical packaging for popular business and personal computer software
- Counterfeiting software are sold at lower prices but they profit because they do not pay designers, developers, programmers, managers etc

SOFTWARE PIRACY – VIOLATION OF SOFTWARE COPYRIGHT

■ Why occurs?

- Copying is easy
- Lack of awareness about copyright act
- Lack of government enforcement
- Do not understand the benefits of original software
- Lack of recognition for IP
- Greedy and careless
- High demand from users
- Problem with cash flow
- Problem with bribery

SOFTWARE PIRACY – VIOLATION OF SOFTWARE COPYRIGHT

- Effect of piracy
 - Reduces competitiveness in software industry
 - Reduces investment in new development of software
 - Users who are using pirated software do not get technical support, assurance, and update
 - Risk of being affected by virus, un-functional, incompatible with hardware
 - Increase in software cost
 - Reduces software development

MORALITY OF COPYING SOFTWARE

- Is it right to copy proprietary software?
- If you violate a licensing agreement, you are breaking the law
- What is the morality of breaking the law?

SOCIAL CONTRACT THEORY

- Assumption : everyone in society ought to bear certain burdens in order to receive certain benefits
- Legal system: to guarantee that people's rights are protected
- Guarantees people will not choose their selfish interests over the common good
- Thus we have *prima facie* obligation to obey the law.
- Therefore, it is wrong to copy software, because copying violates the legal rights of the person or organization owning the copyright.

ARGUMENTS IN SUPPORT PERSONAL COPYING AND COUNTERPOINTS

- *I cannot afford to buy the product*

- There are many things we cannot afford; not being able to afford something is not an excuse for taking it

- *The company is large, wealthy corporation*

- The size and success of the company do not justify taking from it. Programmers and writers lose income too when copying is common

- *Making a copy for a friend is just an act of generosity*

- Surely we have a liberty to be generous, and we can exercise it by buying a copy of the program for the friend

ARGUMENTS IN SUPPORT PERSONAL COPYING AND COUNTERPOINTS

- *This violation is insignificant compared to the billion of dollars lost to piracy by dishonest resellers making big profit*
- Yes, large-scale commercial piracy is worse. That does not imply individual copying is ethical. And, if the practice is widespread, the losses become significant
- *Everyone does it. You would be foolish not to*
- The number of people doing something does not determine whether it is right.

FAIR USE – GROUP DISCUSSION

- An art lecturer takes slide photographs of a number of paintings reproduced in a book about Renaissance artists. She uses the slides in her class lectures
- You have scanned the entire text of Md. Akhir Leman's novel "*Bahagia Sesudah Derita*" (1958) and placed it on Blackboard for your students to use. Is this use okay?
- You have found a website which legally streams a copyrighted version of the title track from Bob Seger's 1976 album "NightMoves". You decide to add a link to this page on your course's Blackboard site so that students can listen to the song. Is this use okay?

...CONTINUE

- You have scanned an advertisement from a recent issue of Time magazine and placed it on your course's Blackboard site for your students to interpret and critique. Is this use likely to be considered a fair use?
- You have scanned a book written by one of your colleagues at another university. She has given you written permission to post the work on your Blackboard site. Is this use permitted under copyright law?

SOLUTION - TECHNOLOGY

- Encoding an expiration date in free sample
- Hardware dongle
- Copy protection scheme
- Digital right management
- A collection of techniques that allow prevention of copying or viewing
- Allow the producer to specify what may be done
- Encrypt digital content
- Digital marking so devices can recognize content as copy-protected

SOLUTION - TECHNOLOGY

- Secure digital music initiative (SDMI)
 - Goals :- create copy-protected CDs, secure digital music downloads
 - Consortium of 200 companies developed “digital watermarking” scheme
 - Failed
- ❖ Internet copying became huge before SDMI ready
- ❖ Some SDMI sponsors were electronics companies
- ❖ Digital watermarking encryption cracked

SOLUTION - TECHNOLOGY

■ Encrypting DVDs

- Contents of DVDs encrypted using Content Scramble System (CSS)
- Need decryption keys to view a DVD
- Jon Johansen wrote a decryption program for Linux
- 2600 Magazine published the code
- Motion picture studios sued 2600 Magazine and won
- Johansen tried in Norway and found not guilty

SOLUTION - TECHNOLOGY

- Making CDs Copyproof
 - Two approaches
 - ✓ Exploit difference between CD-ROM drives in computers and CD players
 - ✓ Encode patterns into audio that translate into annoying sounds when decode
 - Two obstacles
 - ✓ Large community eager to “crack” encryption schemes
 - ✓ People can record what they hear

SOLUTION – MARKETS AND MANAGEMENT

- **Subscribe to services**
- **Collect fees from users and large organization**
- **Meter usage of IP on a network**
- **Offer discounts to educational users**
- **Educate the public about the value of IP belonging to creators and publishers**

SOLUTION - ENFORCEMENT

- Identify abusers and shut them down
- Monitor abuses
- Enforce laws and punish abusers

SOLUTION - ENFORCEMENT

- Digital Millennium Copyright Act
- Extends length of copyright
- Extends copyright protection to music broadcast over internet
- Makes it illegal for anyone to circumvent copy controls, even for fair use purpose

PEER-TO-PEER NETWORKS

- Napsters
- FastTrack
- BitTorrent
- MP3 spoofing
- Universities caught in the middle
- Legal music services on the internet

NAPSTER

- Peer-to-peer music exchange network
- Began operation in 1999
- Sued by RIAA for copyright violations
- Courts ruled in favor of RIAA
- Went off-line in July 2001
- Re-emerged in 2003 as a subscription music service

FASTTRACK

- Second-generation Peer-to-peer network
- Used by KaZaA and Grokster
- Sued by RIAA for copyright violations
- Distributes index among large number of “supernodes”

BITTORRENT

- Broadband connections: download much faster than upload
- BitTorrent speeds downloading
 - Files broken into pieces
 - Different pieces downloaded from different computers
- Used for downloading large files

MP3 SPOOFING

- Companies support efforts to post bad MP3 files on peer-to-peer networks
- Goals: make downloading of MP3 files unreliable
- Peer-to-peer software makers responded by adding user ratings

UNIVERSITIES CAUGHT IN MIDDLE

- Universities hotbed for file sharing
- ✓ High-speed internet access
- ✓ High-capacity file servers
 - *In 2003 RIAA sued four students for about \$100 billion (settle for \$50,000)*
 - *Different university responses*
- Taking PCs of students
- Banning file-sharing
- Signing agreements with legal file-sharing services

LEGAL MUSIC SERVICES ON THE INTERNET

- Subscription services for legal downloading
- Some based on monthly fee; some free
- Consumers pay for each download
- Apple's iTunes Music Store leading service

OPEN-SOURCE SOFTWARE : CONSEQUENCES OF PROPRIETARY SOFTWARE

- Increasingly harsh measures being taken to enforce copyrights
- Copyrights are not serving their purpose of promoting progress
- It is wrong to allow someone to “own” a piece of intellectual property

FREE SOFTWARE

- An idea, an ethic, advocated and supported by a large loose-knit group of programmers who let people copy and modify their software, often without charge, and encourage to do so
- *Free* in free software means freedom, not necessarily lack of cost, though often there is no charge
- Advocate allows unrestricted copying and making source code available(open source)

OPEN SOURCE DEFINITION

- No restrictions preventing others from selling or giving away software
- Source code include in distribution
- No restrictions preventing others from modifying source code
- No restrictions regarding how people can use software
- Same rights apply to everyone receiving redistributions of the software (copyleft)

BENEFICIAL CONSEQUENCES OF OPEN SOURCE

- Gives everyone opportunity to improve program
- Frequent appearance of new version
- Eliminates tension between obeying copyright law and helping others
- Programs belong to entire community
- Shifting focus from manufacturing to service

EXAMPLES OF OPEN SOURCE SOFTWARE



- BIND
- Apache web server
- Sendmail
- Perl
- Python, Ruby, TCL/TK, PHP
- GNU compilers for C, C++, objective-C, fortran, Java, and Ada

ISSUES FOR SOFTWARE DEVELOPERS

- Should you copyright or patent software?
- First aspect: what is the nature of a new program, or a new kind of program? Is it an invention, a new idea? Or is it a “writing,” an expression of ideas, algorithms, techniques?
- Second aspect: what are the practical consequences of each choice in term of encouraging innovation and production of new products?

ISSUES FOR SOFTWARE DEVELOPERS

- Should you copyright or patent software?
- Copyrights :
 - Protect the expression of an idea in a fix and tangible form
 - Are cheap, easy to obtain, and last a long time
 - Allow fair use of the IP

ISSUES FOR SOFTWARE DEVELOPERS

- Should you copyright or patent software?
- Patents :
 - Protect new, non-obvious, and useful processes
 - Are expensive, difficult to obtain, and last for short period of time
 - Allow licensing to other developers

SIMILAR SOFTWARE PRODUCTS

- *Whelan Associates vs Jaslow Dental Laboratory* (1986)
- Court ruled a program that was similar in structure and performance, although written in a different language for a different computer, infringed copyright

➤SIMILAR SOFTWARE PRODUCTS

- *Plain Cotton Co-op Association vs Goodpasture Computer Service (1987)*
- Only literal copying of code was infringement



SIMILAR SOFTWARE PRODUCTS

- *Computer Associates International vs Altai* (1992)
- Court decided by
 - Identify the purpose of the program
 - Remove from consideration the parts that are in the public domain, are common practice, or are the only efficient way of accomplishing some part
 - Compare the remaining parts and see how similar they are
 - Any particular case need expert witnesses and complex analysis of the program